

RNA-Seq Literature Familiarization and Data Characterization

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Group#: 1

Assigned Topics: Heat Stress

Part 3. Understand the Paper Before Downloading Data

Item	Answer
Title of the paper	Transcriptomic Profiling of <i>Paulownia elongata</i> in Response to Heat Stress
Authors	Neerja Katiyar, Niveditha Ramadoss, Dinesh Gupta, Suman B. Pakala, Kerry Cooper, Chhandak Basu
Year	2021
Journal	Plant Gene (Vol. 28, 100330)
Organism	<i>Paulownia elongata</i>
Tissue / organ / cell type	Leaves
Biological question	Which genes and pathways are involved in <i>P. elongata</i> 's transcriptomic response to heat stress, and can they explain its heat tolerance?
Control condition	Plants grown at 25°C, same photoperiod (12h light/12h dark)
Treatment or stress condition	Plants exposed to 40°C for 24 hours (heat stress)
Number of biological replicates	3 per condition (3 control, 3 heat-stressed)
Sequencing platform	Illumina MiSeq (MiSeq Reagent Kit v3)
Single-end or paired-end sequencing	Paired-end
Read length, if reported	76 bp
RNA-seq repository	NCBI SRA / GEO
BioProject / Study accession	PRJNA485845 (Note: SRA record shows PRJNA488054)
Run accession numbers	SRP158908 (SRA study); individual runs include SRR7758332
Reference genome or transcriptome used by the authors	No public reference genome existed for <i>P. elongata</i> ; authors built a de novo transcriptome assembly using Trinity (v2.0.6)
Main RNA-seq software or tools reported by the authors	Trinity (assembly), DIAMOND/BLAST (contamination screening), MEGAN6, OmicsBox/Blast2GO (annotation, GO/KEGG analysis), Salmon (quantification), edgeR (differential expression)

Part 4. Experimental Design Diagram

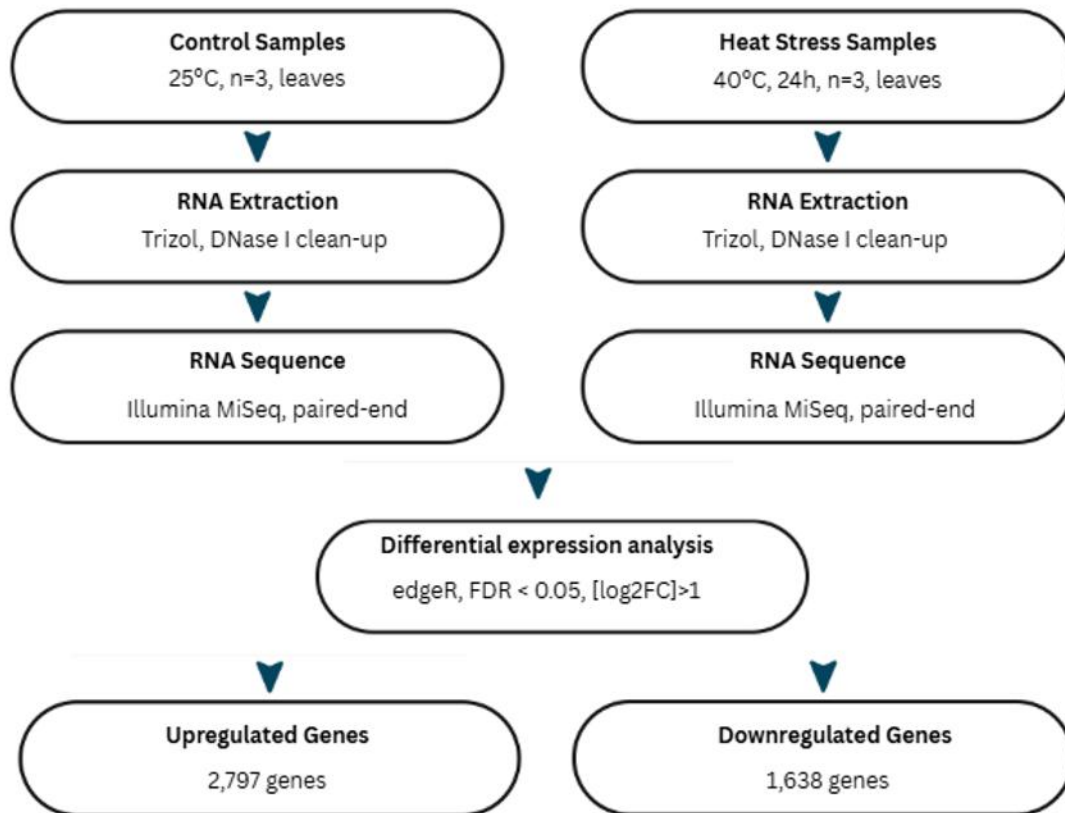


Figure 1. Experimental design based on the Transcriptomic Profiling of *Paulownia elongata* in Response to Heat Stress paper.

- **What was changed or manipulated?**
 - Temperature (40°C for 24 hours vs. 25°C control)
- **What was the control?**
 - Plants grown at 25°C, same photoperiod
- **What was measured?**
 - Genome-wide gene expression (RNA-seq → differential expression)
- **How many biological replicates were used?**
 - 3 per condition (H1–H3 heat stress, C1–C3 control)
- **What tissue or cell type was sequenced?**
 - Leaves

Part 6. My Assigned RNA-Seq Sample

Item	Answer
Run accession number	SRR7758332 (under SRX4614119, GSM3357468, Control_sample2)
Condition (control/treatment)	Control
Biological replicate number, if known	Replicate 2
Single-end or paired-end	Paired-end
Approximate file size, if available	~74.9 Mb (download size); 1,349,099 spots, 203.7M bases

Part 8. Examine the FASTQ File

1. How is a FASTQ file different from the FASTA genome file used in the previous assignments?

- FASTA stores only sequence identifiers and nucleotide sequence; FASTQ adds a per-base quality score line for every read (4 lines/read vs. 2 for FASTA).

2. What information is present in FASTQ that is not present in FASTA?

- FASTQ has base-calling quality scores (Phred scores), absent in FASTA.

3. Why are quality scores important in RNA-seq?

- Quality scores show how confident the sequencer is in each base call. Low-quality bases can introduce false variants/errors into downstream mapping and expression counts, so they need to be checked/trimmed before analysis.

Part 10. Characterized RNA-Seq Dataset

Item	Answer
RNA-seq run accession	SRR7758332 (GSM3357468, Control_sample2)
Condition	Control
Single-end or paired-end	Paired-end (R1 forward, R2 reverse)
Number of reads / sequences	1,349,099 (R1 and R2 each)
Read length	34-76 bp
File size	~74.9 Mb downloaded; R1 and R2 combined = 203.7 Mbp total bases
GC content	41%
General per-base sequence quality	Phred scores mostly in the green zone (~32-38) - PASS
Presence of adapter content	None detected - PASS
Presence of overrepresented sequences	None detected - PASS
Any quality warning observed	Per base sequence content - FAILED; Sequence Length Distribution - WARNING

Part 12. Group Comparison Table

Student	Accession	Condition	Reads	Read length	GC %	Main QC observation
Ferrer	SRR7758332	Control	1,349,099	34-76 bp	41%	Per base seq. content FAILED; Seq. Length WARNING
Alama	SRR7758331	Control	3,147,060	35-76 bp	39%	R1 failed "Per Base Sequence Content"; mild duplication/length-distribution warnings; and trace polyA adapter content.
Bullanday	SRR7758328	Treatment	4,956,271	35-76 bp	45%	Per Base Sequence Content Failed
Calipas	SRR7758330	Treatment	3,441,734	35-76 bp	41% - 42%	Overall quality: per base sequence content failed and sequence length distribution warned

Part 13. Interpretation Questions

1. What biological question was the original RNA-seq study trying to answer?

- What genes/pathways let *P. elongata* tolerate heat stress, since this hasn't been characterized despite the species' known thermotolerance.

2. Why did the authors use RNA-seq instead of only examining the genome?

- The genome alone shows what genes exist, not which are actively transcribed/regulated under a specific condition (heat stress); RNA-seq captures the dynamic expression response.

3. What is the difference between genomic DNA and the RNA molecules measured by RNA-seq?

- DNA is the fixed genetic blueprint (same in all cells); RNA (mRNA) reflects which genes are actively being expressed at a given time/condition; RNA-seq measures this expression, not the static sequence.

4. What is a biological replicate and why is it important?

- An independent biological sample (separate plant/organism) treated identically, used to distinguish true treatment effects from natural biological variation. The paper used 3 replicates per condition.

5. What is the difference between single-end and paired-end sequencing?

- Single-end sequences a fragment from one direction only; paired-end sequences both ends (forward + reverse), giving better mapping accuracy and info on fragment size. This study (and my dataset) is paired-end (R1/R2).

6. What is a FASTQ file?

- A text format storing each read's ID, sequence, a separator, and quality scores (4 lines/read).

7. What information does FastQC provide?

- Per-base and per-sequence quality scores, GC content, sequence length distribution, duplication levels, overrepresented sequences, and adapter content, an overall quality snapshot of raw reads.

8. What does a high per-base quality score indicate?

- Low probability of a base-calling error at that position (Phred $\geq 30 \approx 99.9\%$ accuracy) i.e., trustworthy reads.

9. Why can adapter contamination be a problem?

- Leftover sequencing-adapter sequence in reads can mismatch during alignment or be misread as biological sequence, distorting expression counts.

10. Were all RNA-seq samples in your group similar in quality? Explain.

- Mostly yes, but not identical. All four samples passed per-base and per-sequence quality scores, adapter content, and overrepresented sequences checks, and all shared the same failing flag on per base sequence content (likely from random-hexamer priming bias at the start of reads, common in RNA-seq). However, GC content varied more between conditions than within them: the two control samples (Alama 39%, mine 41%) were close to each other, while the two treatment samples (Bullanday 45%, Calipas 41-42%) ran higher, hinting at an expression shift under heat stress. Read counts also varied widely, from about 1.3M (mine) to nearly 5M (Bullanday's), reflecting differences in sequencing depth rather than quality per se.

11. Did any sample show a possible quality problem? What was it?

- My sample (SRR7758332) showed a Sequence Length Distribution warning in addition to the per base sequence content failure that all group samples shared. This likely reflects the range of trimmed read lengths (34-76 bp) rather than a serious quality issue, since duplication levels, adapter content, and overrepresented sequences all still passed.

12. What additional steps would be needed before the researchers could compare gene expression between control and treatment samples?

- Read trimming/QC filtering, alignment or pseudo-alignment to a reference (or transcriptome, since no genome exists here, hence Trinity assembly), quantification (e.g., Salmon), and differential expression testing (e.g., edgeR), as done in the published paper.

Part 15. Group Summary

1. What study did the group choose and why is it relevant to the assigned topic?

-The group selected Katiyar et al. (2021), "Transcriptomic Profiling of Paulownia elongata in Response to Heat Stress," because it directly addresses the group's assigned topic of heat stress by identifying the genes and pathways behind P. elongata's heat tolerance using RNA-seq.

2. What organism, tissue, control, and treatment were used?

-P. elongata leaves; control plants grown at 25°C; treatment plants exposed to 40°C for 24 hours; 3 biological replicates per condition.

3. How many RNA-seq samples were included in the original study?

-6 total — 3 control (C1–C3) and 3 heat-stressed (H1–H3).

4. What sequencing platform and read type were used?

-Illumina MiSeq (MiSeq Reagent Kit v3), paired-end, 76 bp reads.

5. What did the group learn from examining the raw RNA-seq data?

-The FASTQ files followed the expected four-line structure with per-base Phred quality scores; overall base quality was high (mostly Phred 34–38) across the group's samples, but read depth differed substantially between individually assigned runs, and a recurring "Per Base Sequence Content" failure showed up specifically in the two control replicates.

6. Were there noticeable differences in data quality among samples?

-Yes. Read counts and GC% varied across the four runs, and the two control-condition samples shared the same QC failure pattern, distinct from the treatment samples.

7. What analysis did the original authors perform after quality control?

-The authors built a de novo transcriptome with Trinity (no reference genome existed), screened for contamination with DIAMOND/BLAST and MEGAN6, assessed assembly completeness with BUSCO, annotated transcripts with Blast2GO/OmicsBox, quantified expression with Salmon, and identified differentially expressed genes with edgeR ($FDR < 0.05$, $|\log_2FC| > 1$), followed by GO and KEGG pathway enrichment.

8. What will the group need to learn next in order to reproduce part of the authors' gene-expression analysis?

-Read trimming/adaptor removal, alignment or pseudo-alignment of reads to a reference or assembled transcriptome, transcript quantification (e.g., Salmon), and differential expression testing (e.g., edgeR), the same pipeline steps the original authors used to go from raw reads to their list of upregulated/downregulated genes.

GitHub Repository link:

https://github.com/rhealynalama-bio/RNASeq_Group-1_Heat-Stress/tree/main/FERRER